

UNIT 3

THE HUMAN BODY

Learning objectives:

By the end of this unit, students will be able to:

- Identify and describe the major systems of the human body.
- Explain the functions of each body system and how they contribute to the overall functioning of the human body.
- Understand the connections between body systems.
- Differentiate between voluntary and involuntary muscles and provide examples of each.
- Recognize the importance of keeping the body healthy through proper diet, exercise, rest, and hygiene.

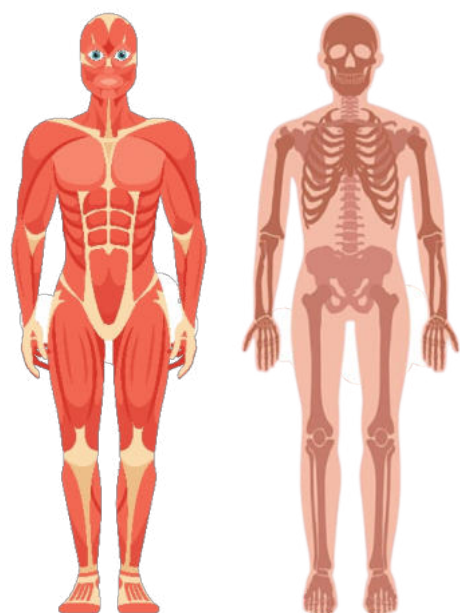
The human body is an amazing machine made up of several systems that work together to keep us alive and healthy. Each system has a unique role, but they all connect to form a powerful team.

OVERVIEW OF BODY SYSTEMS

Here are the major systems of the human body and their main functions:

1. Skeletal System:

- Framework of 206 bones that gives shape, support, and protection.
- Works with muscles to help us move.
- Produces blood cells and stores minerals like calcium.



2. Muscular System:

- Made of muscles that pull on bones to create movement.
- Includes three types of muscles:

Skeletal: Voluntary muscles that help us move.

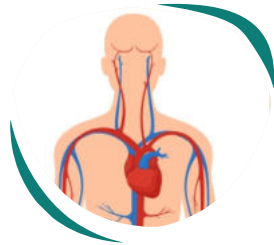
Smooth: Involuntary muscles in organs like the stomach.

Cardiac: Found only in the heart.



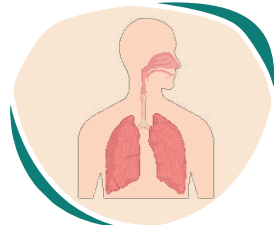
3. Circulatory System:

- Includes the heart, blood, and blood vessels.
- Transports oxygen, nutrients, and waste throughout the body.
- Keeps muscles and organs working by delivering energy.



4. Respiratory System:

- Includes the nose, windpipe, and lungs.
- Brings oxygen into the body and removes carbon dioxide.



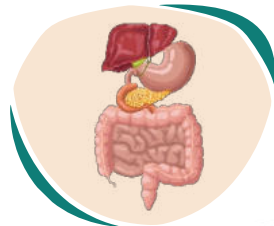
5. Nervous System:

- The control centre of the body, made up of the brain, spinal cord, and nerves.
- Sends and receives signals to control movement, reflexes, and body functions.



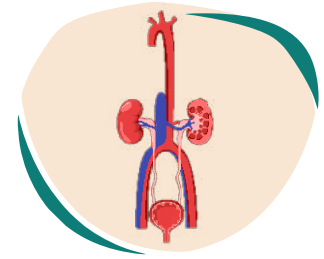
6. Digestive System:

- It helps in digestion.
- Includes organs like the stomach, intestines, and liver.



7. Excretory System:

- Removes waste from the body.
- Includes the kidneys, bladder, and skin.



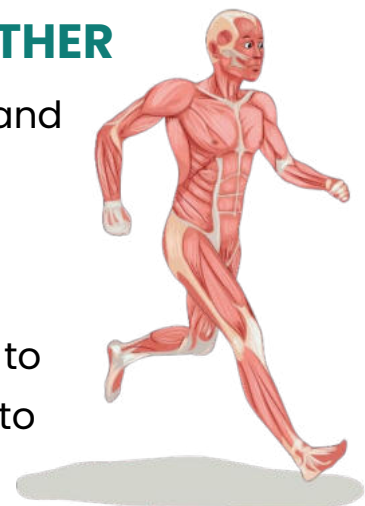
8. Immune System:

- Protects the body from harmful germs and diseases.
- Includes white blood cells and organs like the spleen and lymph nodes.



HOW BODY SYSTEMS WORK TOGETHER

- The systems of the body are connected and depend on each other:
- The **muscular system** works with the **skeletal system** to create movement.
- The **respiratory system** supplies oxygen to the **circulatory system**, which delivers it to the muscles and brain.
- The **nervous system** controls all systems, sending signals for movement, breathing, and digestion.



KEEPING OUR BODY SYSTEMS HEALTHY

- **Diet:** Eat healthy foods like fruits, vegetables, and proteins to strengthen bones, muscles, and organs.
- **Exercise:** Regular activities like running, swimming, or cycling keep muscles, the heart, and lungs strong.
- **Rest:** Adequate sleep allows the body to repair and recharge.
- **Hygiene:** Keep your body clean to prevent infections.





Fill in the blanks.

1. The _____ system gives shape and support to the body.
2. The _____ system pumps blood to deliver oxygen and nutrients.
3. The _____ system removes waste from the body.
4. The _____ system breaks down food into nutrients.
5. The _____ system controls the body by sending signals.
6. _____ muscles are found in the heart.
7. Bones store minerals like _____ .
8. The _____ system protects us from diseases.



Write “T” for True or “F” for False.

1. The circulatory system includes the brain.

2. Muscles and bones work together to create movement.

3. The lungs are part of the respiratory system.

4. White blood cells help fight diseases.

5. The nervous system sends signals to the body.

6. The stomach is part of the skeletal system.

7. The excretory system helps remove waste from the body.

8. Cardiac muscles are voluntary muscles.



Multiple Choice Questions (MCQs).

1. Which system controls the body's movements?

- Circulatory System
 - Nervous System
 - Digestive System
-

2. What is the main function of the respiratory system?

- To break down food.
 - To bring oxygen into the body.
 - To remove waste.
-

3. Which organ is part of the digestive system?

- Heart
 - Lungs
 - Stomach
-

4. Which system removes waste from the body?

- Immune System
 - Excretory System
 - Skeletal System
-





5. What does the skeletal system do?

- Pumps blood.
 - Gives shape and protects organs.
 - Removes waste.
-

6. What is the role of the immune system?

- Protect the body from germs.
 - Break down food.
 - Supply oxygen to muscles.
-

7. Which muscle type is found only in the heart?

- Smooth Muscle
 - Skeletal Muscle
 - Cardiac Muscle
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8. How do the respiratory and circulatory systems work together?

- They move bones and muscles.
 - They provide oxygen to the body.
 - They remove waste from body.
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Answer the following questions.

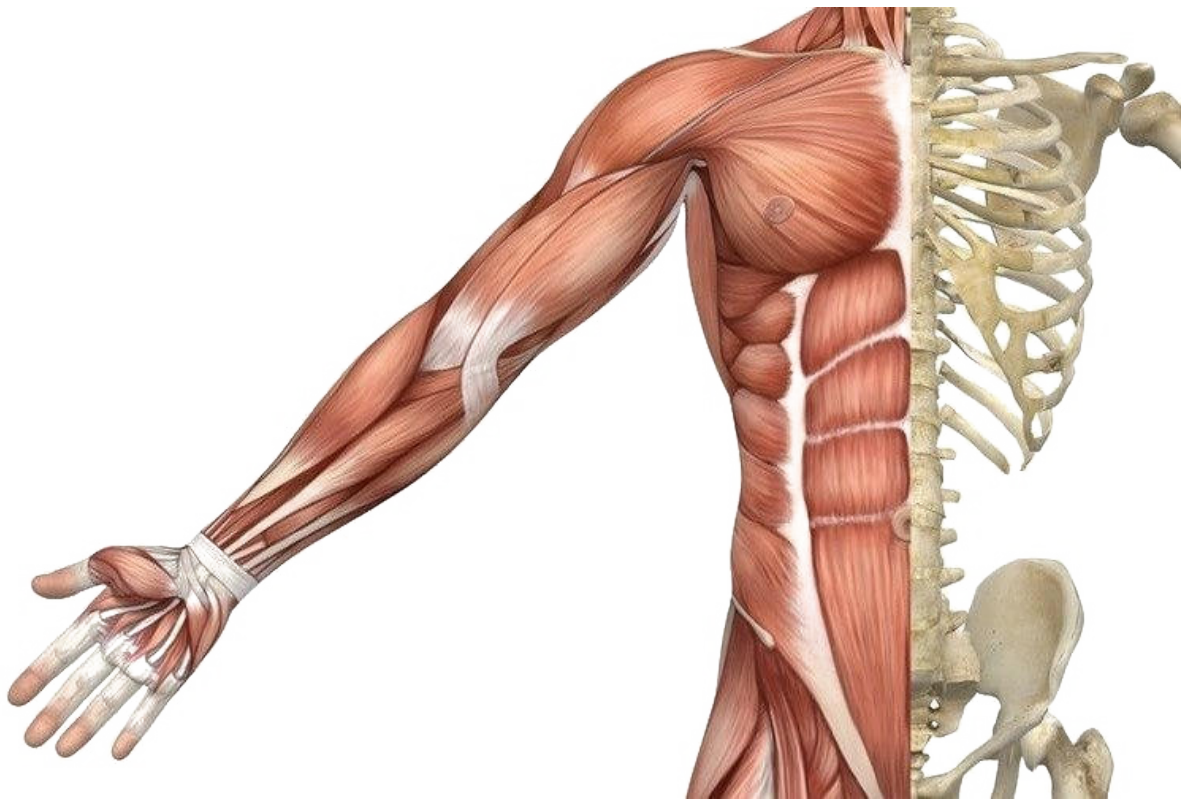
Q1: Name three systems of the human body and their functions.

Q2: How do the skeletal and muscular systems work together?

Q3: Why is the circulatory system important?

Q4: What is the main function of the nervous system?

Q5: How can we keep our body systems healthy?





Build a Model of a Working Arm



ACTIVITY

Build a simple model of a working arm using materials provided by the teacher. Show how muscles (rubber bands) pull on bones (cardboard) to create movement at the joint (paper fastener).

Explain how this demonstrates the connection between the skeletal and muscular systems.

